Information for the participant and
Electronic informed consent form

Title of the study: Healthy [Brain] Ageing – a European online project to identify and longitudinally follow-up individuals at risk for neurodegenerative disorders, including Parkinson's disease

Study acronym: HeBA

Co-Sponsors of the study:

University of Luxembourg (UL)/Luxembourg Centre for Systems Biomedicine (LCSB)

Luxembourg Institute of Health (LIH)

Chief Scientific Investigator of the study: Prof. Dr. Rejko KRÜGER

Chief Medical Investigator of the study: Prof. Dr. Rejko KRÜGER

Contact details:

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1 INTRODUCTION

You are invited to take part in a clinical study. The purpose of this document is to provide you with some information on the study to help you decide whether you would like to take part. Your participation is entirely voluntary. If you choose to take part, you may withdraw at any time without having to give any reason. In Luxembourg, this study was authorized by the Ministry of Health and received a favorable opinion from the National Research Ethics Committee (CNER) on 23/08/2022. However, you should not take this latter information as an incentive to participate in this study.

2 WHAT IS THE PURPOSE OF THE STUDY?

2.1 About age-related neurodegenerative diseases

2.1.1 What are age-related neurodegenerative diseases?

Age-related neurodegenerative diseases are progressive diseases of the nerves in the brain that occur more frequently with increasing age. The most common age-related diseases are dementia, Alzheimer's disease, and Parkinson's disease. It is estimated that 55 million people worldwide are affected by Alzheimer's disease or another form of dementia, and 8.5 million by Parkinson's disease. With the higher life expectancy, this number will increase significantly in the coming years.

Most people affected by age-related neurodegenerative diseases are older than 60, although younger people could also be affected.

Depending on the disease, various mechanisms can lead to the premature aging of nerve cells in the brain, whereby the cells "infect each other," thus leading to the disease progression.

Common symptoms are memory problems, changes in character, general slowing down, stiffness and tremor. There is no cure for most age-related neurodegenerative diseases, and the triggers are often unknown.

2.1.2 How are age-related neurodegenerative diseases diagnosed?

At present, the diagnosis is primarily made clinically, i.e., by a doctor's consultation and a physical medical examination. Depending on the complaints, additional examinations, such as a detailed psychological test or a test of mobility, can be helpful.

Often, imaging of the head and brain is done, for example, an MRI or computer tomography (a so-called slice X-ray), especially in unclear cases or to rule out other treatable causes of the symptoms.

Early signs of age-related neurodegenerative diseases can be cognitive deficits (memory disorders, disorientation, change of character) and movement disorders (stiffness, tremor, slowing down, changes in writing, or decreasing facial expressions). Other symptoms can be, for example, a smell disorder, depression, constipation, or sleep disorders.

2.1.3 Why do we want to diagnose age-related neurodegenerative diseases at an early stage?

One of the reasons there is no cure for most age-related neurodegenerative diseases is the time of diagnosis. At this stage, a large proportion of the brain's nerve cells have already been lost. In Parkinson's disease, for example, 50-60% of the important dopamine-producing cells have already lost their function by the time of diagnosis. The progressive loss of neurons already begins many years before the first symptoms appear.

Unfortunately, we know little about this very early phase of the disease. If we could detect affected individuals early, we would be a small step closer to our research aim of slowing, stopping, or even reversing the disease process.

2.1.4 How might we identify people at increased risk for age-related neurodegenerative disease before symptoms appear?

Previous research indicates that anxiety, depression, forgetfulness, personality changes, sleep disturbances, constipation, or olfactory disorders may occur up to 20 years before the disease can be diagnosed. Many people above 60 will likely have one or another of these symptoms without developing a neurodegenerative disease later. However, we could identify individuals at exceptionally high risk for age-related neurodegenerative diseases by identifying people exhibiting several of these symptoms. This is precisely our goal.

2.2 About the Healthy Aging study

Healthy Aging (HeBA) is a unique research study that aims to identify people at high risk of age-related neurodegenerative disease before diseases such as dementia or movement disorders.

HeBA is a European multicenter collaborative study in Luxembourg, Kassel in Germany, Innsbruck in Austria, Barcelona in Spain, and London in the United Kingdom. The partners will follow a harmonized study protocol to allow comparison but adapt to each site-specific need. The information in this document is adapted to Luxembourg.

We use a combination of simple questionnaires that are asked online via computer. These questions capture factors associated with an increased risk of developing an age-related disease. We want to find out whether the overall result of these tests can tell us anything about the risk of developing an age-related disease.

2.2.1 Will this study identify people who will develop an age-related neurodegenerative disease?

No. This study aims to optimize the assessment of risk determinants for certain neurodegenerative diseases, it is not designed as a tool to predict the evolution of the state of health of each participant. Although there is already knowledge about the very early stages of these age-related diseases, it is currently difficult to determine precisely who will develop a neurodegenerative disease later. The results of this study will help develop more reliable methods to identify people at increased risk.

2.2.2 What is unique about the Healthy Aging study?

HeBA is innovative because it is almost entirely online. This means that participants don't have to leave their homes to participate, and we can evaluate information quickly, cheaply, and efficiently.

We hope that in this way, more people will be able to participate in the study and possibly benefit from the results in the future.

HeBA is also unique because it targets the earliest known symptoms of age-related neurodegenerative diseases.

3 HOW WILL THE STUDY BE CONDUCTED?

The study has two parts. The first part is the online questionnaire and smell test. Based on the online questionnaire answers, the study team will contact a subset of participants to complete a smell test they will receive at home.

The second part consists of an in-person visit. Some participants completing the smell test will be invited for detailed in-person clinical examinations.

You can choose to participate only in the first phase (online questionnaire and smell test) or, if invited, join the following study step of an in-person visit. Two online follow-up questionnaires will be sent at least four weeks apart to all participants who agreed to be contacted again (in phase 1). You are free to complete only one or none of them.

3.1 Who can participate in the study?

The study in Luxembourg is looking for people between 50 and 80 years old with internet access and NOT diagnosed with age-related neurodegenerative diseases who reside in Luxembourg or the Greater Region (Belgium: Wallonia, the German-speaking Community; France: Lorraine; and Germany: Saarland and Rhineland-Palatinate).

People who have family members with neurodegenerative diseases are welcome to attend.

3.2 Who cannot participate in the study?

1) People with a known diagnosis of age-related neurodegenerative diseases such as Parkinson's or Alzheimer's disease or other neurological movement disorders, dementia, stroke, or amyotrophic lateral sclerosis (ALS).
2) Some medications can aggravate or trigger symptoms of age-related neurodegenerative disease. Therefore, we ask for your medication (please have it ready).
3) People who do not live in Luxembourg or the Greater Region.

3.3 What will happen if I participate in the study?

After agreeing to participate in the study by filling out the electronic Consent Form on the secure website of the HeBA study, you will be asked to provide your date of birth and contact details (first name, surname, mobile phone number, e-mail, postal address). Your personal identification information will be used to re-contact you in the context of this study if you qualify further.

You will then be asked to complete the online questionnaire, which asks for various information about medical concerns and life circumstances that may influence the risk of age-related diseases, particularly questions on:

- Demographics and socio-economics
- Psychology
- Quality of life
- Cognition
- Sleep

The questionnaire will take you up to 45 minutes to complete. Since you will be asked about your medical conditions and medications, please have this information at hand.

3.4 Will the study team contact me after completing the online tests?

The study team might contact you, but for most participants, no further contact will be required. We aim to contact up to 3000 participants by post to send a home olfactory test. The smell test will be sent to qualifying participants based on the answers to the online questionnaire.

However, if you are contacted, it does not mean that you are at increased risk, as we will contact participants with different risk scores (low- and high-risk).

Based on the smell test results, you may be invited to present yourself at the clinic for a medical examination and other tests (see below). Beyond that, no further contact will be made by our study team. The initial analyses of the overall data (from all participants) provided promising results, and therefore as indicated previously we decided to continue the study. We therefore ask you to fill two additional questionnaires. The first one is about your hypothetical preference to receive information about your personal risk of developing a neurodegenerative disorder and some additional questions about personality traits and neuropsychological profiles. In a second questionnaire we ask you to repeat the initial online survey to help the study team monitor potential changes in your health and lifestyle over time. Of course, you can refuse this at any time.

3.5 How long will I participate in the study?

For most participants, the study is completed after filling out the online survey.

The study is planned to run for ten years, although after the online phase, investigations of selected participants will take longer.

Some participants may be asked to retake the online test annually for a longitudinal evaluation, allowing the study team to assess changes over time.

3.6 What happens if I am one of the participants invited for further examinations?

Participants invited to perform a smell test and/or undergo a medical examination will be contacted after an initial evaluation of the data. Participation is voluntary. If you are invited for an appointment at the research clinic in Strassen, you will be invited to perform a clinical examination and donate samples.

- Medical interview (with a new explanation and declaration of consent for this part of the study)

- Medical examination
- Olfactory and memory tests
- Blood
- Saliva
- Urine

You will have the possibility, if you wish, to donate the following samples for the research as well:

- Stool sample (we will give you a dedicated collection kit to take home, with the help of which you can collect a small amount of stool and send it to us by mail)

For this part of the HeBA study, the clinical examination and sample collection in Luxembourg will be in the context of the related established Biomaterial collection for neurodegenerative disease research (ND Collection), approved by the Ministry of Health and with a favorable opinion from the National Research Ethics Committee (CNER). This national research collection was established in 2015 by the Integrated BioBank of Luxembourg (IBBL), part of the Luxembourg Institute of Health (LIH), and the Luxembourg Centre for Systems and Biomedicine (LCSB), part of the University of Luxembourg (UL) in the context of the National Centre of Excellence in Research on Parkinson's Disease (NCER-PD; https://www.parkinson.lu). The ND Collection aims to collect clinical data and samples from individuals with and without neurodegenerative disease to answer the most urgent research questions, such as defining disease markers and studying mechanisms causing premature neuronal aging.

The clinical examination will, of course, take place after separate clarification and consent. The in-person examinations will take about one morning. You are fully free to refuse specific individual tests.

4 USE OF MY DATA

You are invited (i) to share data by completing a questionnaire for this study (primary use) and (ii) to authorize the subsequent use (secondary use) of data concerning you (except for your direct identifying data) for the purposes of future medical research in the field of neurodegenerative diseases. An information document and informed consent form dedicated to the secondary use of your data will be provided to you separately.

5 WHAT ARE THE POSSIBLE RISKS AND SIDE EFFECTS?

There are no anticipated risks of participation. Participants invited for in-person visits will be asked to provide blood, urine, saliva, and stool samples. There are minimal risks associated with a blood draw. HeBA is designed as a non-invasive study; however, we feel that the minimal risk of blood sampling is justified compared to the potentially high informative value of the analyses. If you are one of the few participants in this second part of the study, we will inform you about these and the following aspects. Blood draws may cause mild pain and have a minimal risk of bleeding, bruising, nerve damage, or infection (<1% of participants). Blood collection is performed by experienced personnel only. The collection of stool, urine, and saliva is done by yourself and is risk-free.

6 WHAT ARE THE BENEFITS OF TAKING PART IN THE STUDY?

You will not benefit directly by participating in this study. Your participation is voluntary. You will receive no compensation for your involvement or any subsequent developments resulting from the study.

However, your participation is important because you will provide information to improve research on neurodegenerative diseases, such as finding ways to diagnose age-related neurodegenerative diseases at the

earliest possible stage. This would potentially pave the way for better treatments and cures, benefiting people worldwide with age-related neurodegenerative diseases.

7 CONFIDENTIALITY AND PROTECTION OF PERSONAL DATA

Please read the information section at the end of this document (pages 9 to 11) describing how we will process your personal data as part of this study.

8 COSTS ASSOCIATED WITH YOUR PARTICIPATION

There are no costs for you to participate in the study.

9 INSURANCE

The data collected during this study is insured against data breaches through the University of Luxembourg. Since this is an online-only study, no additional insurance against accidents is applicable. Participants joining the in-person part of the study will be covered against potential damages through a separate insurance policy.

10 YOUR DECISION TO PARTICIPATE

Your decision to accept or decline to participate in this study will in no way influence the quality of the care you are given. If you choose to take part, you may withdraw at any time without providing a reason.

Before you participate in the study, you will need to provide your consent by completing the electronic consent form. The Chief Medical Investigator will keep a copy of this document in a safe place, and you can download a copy of the document from the study website. Should you have any questions about this study, you can contact the Chief Medical Investigator of the study or a member of the study's clinical team at any time.

If you decide to participate in this clinical study, we ask that you:

- Give your full cooperation throughout the study.
- Do not conceal any information on the state of your health, the medication you take, or any symptoms you might experience.

If you want further information on the study, please contact +352 621 789 769 (during working hours: 8h-17h).

Electronic informed consent form

- I declare that I have read and understood the information provided in the participant information.
- I have had enough time to consider my involvement in the study and discuss it with a person of my choice, such as my general practitioner or a family member. I have had the opportunity to ask all questions that have occurred to me related to the study, and I have received satisfactory responses.
- I am aware of what is expected of me as a participant in this study.
- I am aware that the study consists of two parts and that I may be contacted again for further assessments. I am aware that I can also only participate in the first part (the online questionnaire and the smell test).
- I am aware that my participation in this study is entirely voluntary and that I am free to withdraw at any time without giving a reason for my decision and without being held liable for any material or non-material damages. I will only need to inform the investigating doctor or the clinical team of my decision.
- I accept that the results from this study may be disclosed and reported in scientific publications. The way in which these results are presented will in no way enable me to be identified, either directly or indirectly.
- As described in the information document on the processing of my personal data as part of this study, I understand that any personal information gathered in relation to this study will be treated as strictly confidential, in accordance with the General Data Protection Regulation (EU) 2016/679 of 27 April 2016 (known as the GDPR) and all subsequent texts replacing or supplementing this Regulation (in particular, Luxembourg's law of 1 August 2018 on the organization of its National Commission for Data Protection and the implementation of the GDPR).
- I have a copy of the present document, together with the information document explaining how my personal data will be processed as part of this study. I understand that I have the possibility to download a copy of the present document from the study website.

In the table below, if you answer NO to any of the statements followed by an asterisk (*), you will not be able to participate in the study, as these are essential to its objectives.

	YES	**NO**
I willingly agree to take part in this study under the terms set out on the attached information sheet. (*)	☐	☐
Should this study be continued in the future to investigate changes over time, I agree to be contacted again by the person leading the study or his delegated clinical team to participate further.	☐	☐
I agree to be contacted again by the person leading the study or his delegated clinical team to take part in **another study**.	☐	☐

First name and surname of participant:
Sex at birth (male/female/other):
Postal address:
E-mail address:
Telephone number:

Read below information for the participant on the processing of personal data

Data Protection Notice

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Professor for Clinical and Experimental Neuroscience, University of Luxembourg, Director of Transversal Translational Medicine, Luxembourg Institute of Health, and neurologist at Centre Hospitalier de Luxembourg

Phone number: +352 621 789 769 (during working hours: 8h-17h)

E-mail address: heba@uni.lu

CONFIDENTIALITY AND PROTECTION OF PERSONAL DATA

As part of the study entitled Healthy Aging (the Study), we will need to collect and process some of your personal data to enable us to meet the scientific objectives of the Study.

The UL/LCSB and LIH are jointly responsible for collecting, analyzing, and, more generally, processing your personal data, and for ensuring that it is protected, in accordance with the General Data Protection Regulation (EU) 2016/679 of 27 April 2016 **(known as the GDPR)** and all subsequent texts replacing or supplementing this Regulation, in particular, Luxembourg's law of 1 August 2018 on the organization of its National Commission for Data Protection and the implementation of the GDPR (collectively known as the **Law on Data Protection**).

What type of data will we collect?

If you choose to participate in the Study, we will only collect the personal data we need to achieve the scientific aims of our Study.

We will collect the following data from you using questionnaires in electronic format:

- Directly identifying data: Name, first name, your postal address, e-mail address, telephone number
- Other categories data: age or month/year of birth, sex, your lifestyle, your level of education, your socio-professional group, your working life

- Data relating to your health: comorbidities and current medication for research purposes. Your family medical background relevant for the Study. Your alcohol/tobacco consumption, psychological questionnaires, etc. (as described in the information sheet above).

On what legal basis will we process your data?

The use of your personal data is necessary to enable us to achieve the aims of the study, which we are conducting in the public interest and for the purposes of scientific research (art. 6.1e and art. 9.2j of the GDPR – General Data Protection Regulation 2016/679 of 27 April 2016, applicable as of 25 May 2018).

In practice, if you decide to withdraw from the study, any data collected until your withdrawal will still be used for scientific or statistical research by UL/LCSB and LIH or other researchers to enable the progress of biomedical research, but only in a coded or an anonymous way, and only if you have also agreed to the secondary use of your data. Nonetheless, UL/LCSB and LIH will not collect new data pertaining to you within this study and will delete your nominative data.

Who will have access to your data?

Apart from the Study's investigating doctor and the authorized members of his team working under his responsibility, only the following categories of persons will be able to access your data in relation to their respective duties:

<u>Names or data that directly identifies you (first name, last name, address, telephone number, etc.):</u>

- The Study's clinical team will have access to the Study's correspondence table, linking your identity data to the code assigned to you for the Study.
- The provider of the LamaPoll online survey platfor: Lamano GmBH & Co. KG, located in Berlin, Germany.

<u>Coded (pseudonymized) data</u>:

- The chief scientific investigator and the scientific team working under his responsibility.
- The other Study partners (Paracelsus-Elena-Klinik, Kassel and University Medical Center Göttingen, Department of Neurology at Innsbruck Medical University in Innsbruck, Hospital Clinic de Barcelona, Queen Mary University of London).
- Other external researchers or research bodies, in the private or public sector, as required to meet the needs of the Study or for future scientific research purposes (under no circumstances will we give them data that would reveal your identity).

We may also need to grant access to your coded (pseudonymized) data to service providers carrying out services on our behalf. These organizations are bound by stringent security and confidentiality obligations, as required by the law, and the services they provide are subject to contractual agreements. Finally, in cases where particular inspections or audits need to be carried out, the competent authorities may also have access to your pseudonymized personal data, if necessary, to control the quality of data.

Your data will not be used for any fully automated decision-making processes or any profiling purposes.

Who is responsible for the processing of your personal data?

The Joint Data Controllers in respect of the processing of your data are the Luxembourg Institute of Health, a Public Research Institution having its registered office at 1A-B, rue Thomas Edison, L-1445 Strassen and the

Université du Luxembourg, a Public Institution of Higher Education and Research, having its registered office at 2 avenue de l'Université, L-4365 Esch-sur-Alzette, Luxembourg, acting for its **Luxembourg Centre for Systems Biomedicine**, having its seat at 6, avenue du Swing L-4367 Belvaux.

For any request concerning the processing of your personal data, you can contact the UL/LCSB Data Protection Officer by e-mail at dpo@uni.lu or by post at the following address:

UNIVERSITÉ DU LUXEMBOURG
Data Protection Officer
Maison du Savoir
2, Avenue de l'Université
L-4365 Esch-sur-Alzette

What are your rights?

You will have the right to access and rectify your personal data. In some instances (according to the conditions set out by law*), you have the right to object to how your data is used, to request the deletion of your data, to ask to restrict certain aspects of the processing of your data, and to retrieve your data to forward it to a third party (right of data portability). If you wish to exercise your rights, you may contact the Data Protection Officer of the University of Luxembourg through the contact details listed above. You also have the right to lodge a complaint with Luxembourg's National Commission for Data Protection (CNPD) about the processing of your personal data. Further information is provided on https://cnpd.public.lu/en/particuliers/faire-valoir.html

For information concerning the processing of your personal data as a result of this Study, you can contact the Data Protection Officer (information above, "Who is responsible for the processing of your personal data).

* The University will examine all your requests to exercise rights in accordance with the GDPR and the Luxembourg law of August 1, 2018, on the organization of its National Commission for Data Protection and the implementation of the GDPR. As a result of this assessment and to the extent that we put in place appropriate measures with regard to the protection of your data, we may have to refuse your request if the fulfillment of your request makes it impossible or seriously hinders the achievement of the specific purposes of our study. We will inform you of our decision within one month upon receipt of your request. If the University decides not to grant your request, you have the possibility to contest our decision with the CNPD.

How do we protect your personal data?

The UL/LCSB, in charge of the collection and storage of your electronic data, implements appropriate security measures based on the sensitivity of the information concerned to protect your data from the risk of unauthorized access, loss, fraudulent use, disclosure, alteration, or destruction.

Your data will be treated as strictly confidential. It will also be pseudonymized, meaning that your name will be replaced by a confidential reference code. This code will not directly identify you and will only be used to process your data for scientific purposes. Your identity will never be disclosed in any document produced for the public or other institutions. The correspondence table showing the link between your name and the reference code will be kept confidential and secure by the Chief Medical Investigator and will be stored separately from the other data associated with the Study. Only a small number of people will have access to it (see "Who will have access to your data?" above).

The UL/LCSB also applies the principle of data segregation, i.e., identification data on the one hand and research data on the other hand, are stored on different secure servers to minimize the potential risks of re-identification. Despite all security efforts, the risk of a data breach is not zero but can be described as very low.

For how long will your data be stored?

Your data will be retained, in compliance with legal provisions, for a period of 10 years from the end of the study, scheduled for July 2030. During this period, your data will be pseudonymised when they are used for research; this means that a code replacing your first name and last name will be used. After these 10 years, your personal data will be destroyed. Any additional data we produce from the analysis of your data will be stored in anonymised form, in compliance with legal provisions, for a period of 25 years from the end of the study. Any extension of this retention period will be subject to a request for a favourable opinion to CNER.

Transfers of data outside the European Union.

Your data may be transferred outside the European Union, but only if required to carry out research or make use of the results from the study. Only anonymous data or data that does not directly identify you (coded or pseudonymized data) will be transferred outside the European Union.

Some countries outside the European Union or European Economic Area may not offer the same level of privacy protection as your own country. If this is the case, the UL/LCSB will take appropriate measures to guarantee the protection of your personal data (for example, by including standard contractual clauses on data protection in its contracts, adhering to codes of conduct, complying with a particular certification mechanism, or acting in accordance with your explicit consent).

For more detailed information on the appropriate measures implemented by the UL/LCSB, you can send a request to the Data Protection Officer of the University of Luxembourg by e-mail to dpo@uni.lu.